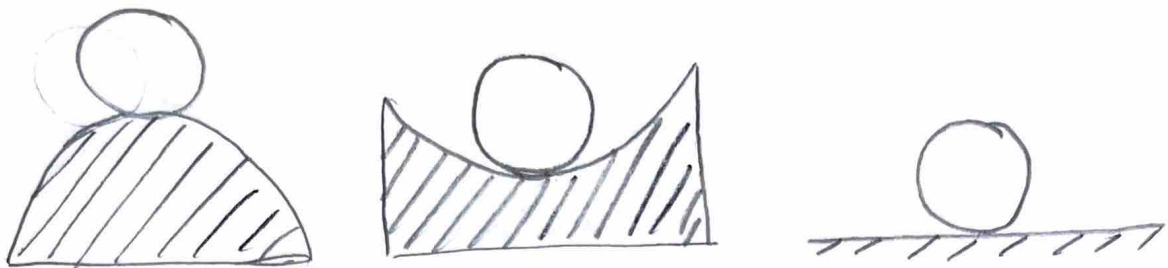
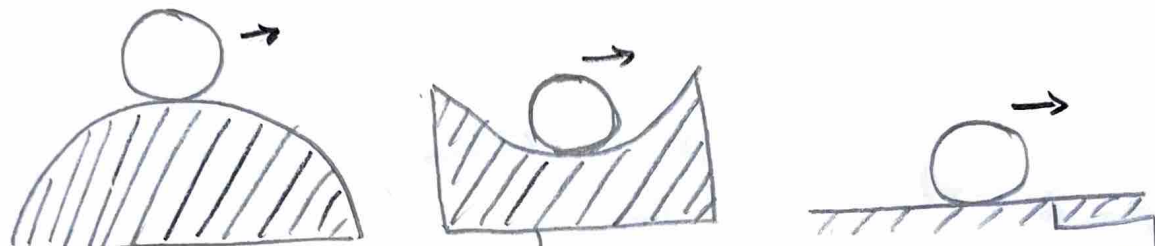


BUCKLING

CONCEPT OF STABILITY :



UNSTABLE " STABLE " NEUTRALLY STABLE
 BASE STATE
 [INITIAL STATE OF EQUILIBRIUM]



[APPLY A SMALL DISTURBING FORCE]
 → WHAT HAPPENS IN EACH SYSTEM?

UNSTABLE
 ⇒ Ball rolls off the hill
 ⇒ Cannot regain original state

STABLE
 ⇒ Ball regains its original state, asymptotically

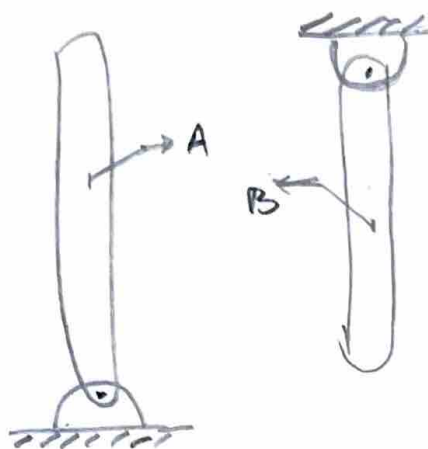
NEUTRALLY STABLE

CAUSE FOR CONCERN

May easily degenerate into instability with slight imperfections such as slightly inclined surface

⇒ Ball does not regain its original state, but may not move far off from original state.

Another example:



"BASE STATES"

→ APPLY A 'SMALL' DISTURBING FORCE

→ WHAT DO YOU EXPECT?

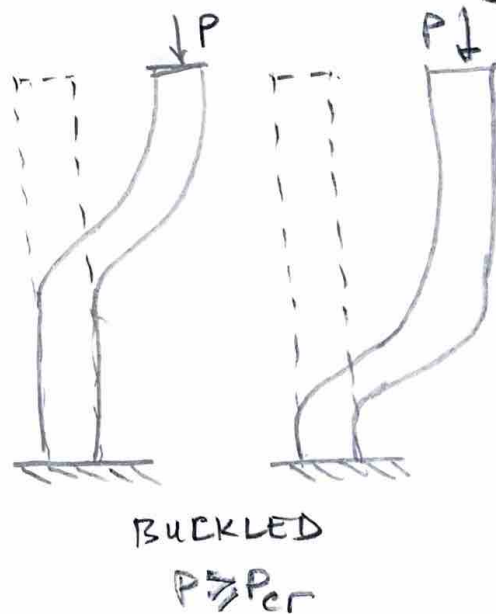
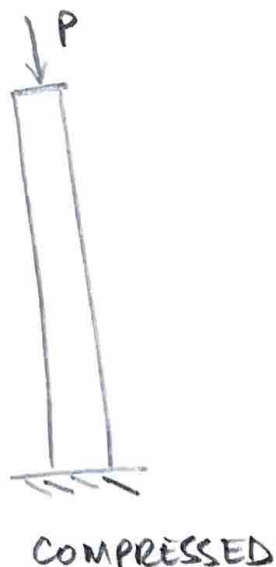
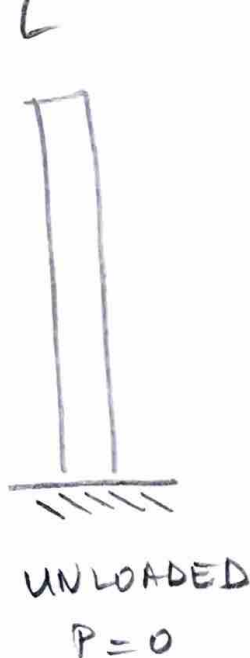
→ A IS UNSTABLE

→ B IS STABLE

MECHANICAL STABILITY: → Ability to resist a small disturbance.

GENERAL CASE OF COLUMN BUCKLING:

{ Classic case of stability in Engineering mechanics }



Consider a straight buckled column:

↳ with LEHI material response

↳ Axial load P is applied through centroid

Case (1): Column compresses as an axial rod

$$\sigma_{xx} = -P/A$$

↳ Displacement

$$u_x(x) - u(0) = \int_0^x -\frac{P}{AE} dx = -\frac{P}{AE} x$$

as long as

$$|P| < |P_{cr}|$$

↳ critical buckling load

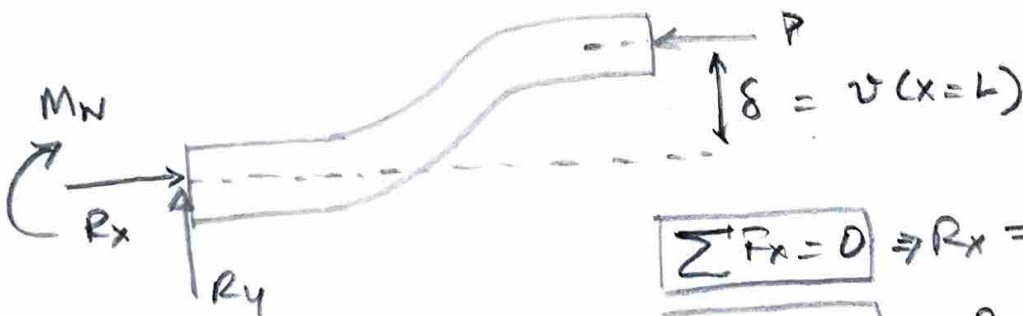
Case (2): $|P| = |P_{cr}|$

↳ Beam appears "bent"

↳ "Bending" due to compressive load

↳ NOT due to transverse load

Apply Equilibrium of the whole (Buckled) shape:

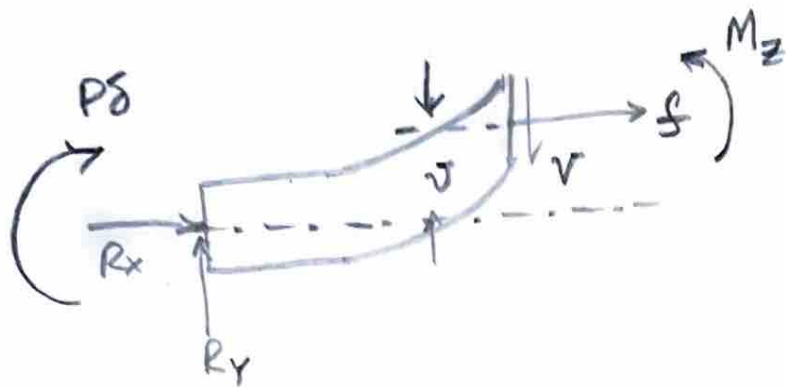


$$\sum F_x = 0 \Rightarrow R_x = P$$

$$\sum F_y = 0 \Rightarrow R_y = 0$$

$$\sum M_z = 0 \Rightarrow P\delta - M_W = 0 \Rightarrow M_W = P\delta$$

Equilibrium of a part of buckled beam:



$$\boxed{\sum F_x = 0}$$

$$\begin{aligned} R_x + f &= 0 \\ \Rightarrow P + f &= 0 \\ \Rightarrow f &= -P \end{aligned}$$

$$\boxed{\sum F_y = 0}$$

$$R_y - V = 0 \Rightarrow V = 0$$

$$\boxed{\sum M_z)_A = 0}$$

$$\begin{aligned} \Rightarrow M_z - f v - P \delta &= 0 \\ \Rightarrow M_z &= f v + P \delta \\ \Rightarrow M_z &= P \delta - P v \end{aligned}$$

Apply moment-curvature relation:

$$\boxed{EI_{zz} \frac{d^2 v}{dx^2} = M_z}$$

$\Rightarrow$ Assume this holds in case of bending

$$\Rightarrow EI_{zz} \frac{d^2 v}{dx^2} = P \delta - P v$$

$$\Rightarrow \frac{d^2 v}{dx^2} + \frac{P}{EI_{zz}} v = \frac{P}{EI_{zz}} \delta$$

$\Rightarrow$ 2nd order
 $\Rightarrow$ Linear
 $\Rightarrow$ Non-homogeneous
OD equation

$\underbrace{\hspace{1cm}}$
constant

$$\left\{ \text{Let } k^2 = \frac{P}{EI_{zz}} \right\}$$

$$\Rightarrow \boxed{\frac{d^2 v}{dx^2} + k^2 v = k^2 \delta} \rightarrow \textcircled{1}$$

Eq ① is a linear equation

∴ Solution:
$$\underbrace{v(x)}_{\substack{\text{Full} \\ \text{solution}}} = \underbrace{v_h(x)}_{\substack{\text{homogenous} \\ \text{solution}}} + \underbrace{v_p(x)}_{\substack{\text{particular} \\ \text{solution}}}$$

① Homogenous solution: From ①

$$\boxed{\frac{d^2 v_h}{dx^2} + k^2 v_h = 0} \Rightarrow \boxed{(D^2 + k^2) v_h = 0}$$

Homogenous eqn. $\rightarrow$ ②

solution to Eq ② is of the form

$$\boxed{v_h(x) = C_1 \cos kx + C_2 \sin kx}$$

$D = \pm ki$

Verify: $\frac{dv_h(x)}{dx} = -kC_1 \sin kx + kC_2 \cos kx$

$$\begin{aligned} \frac{d^2 v_h(x)}{dx^2} &= -k^2 C_1 \cos kx - k^2 C_2 \sin kx \\ &= -k^2 (C_1 \cos kx + C_2 \sin kx) \\ &= -k^2 v_h \end{aligned}$$

$$\Rightarrow -k^2 v_h + k^2 v_h = 0 \quad \checkmark$$

② Particular solution:

$$\boxed{\frac{d^2 v_p}{dx^2} + k^2 v_p = k^2 \delta} \rightarrow \text{③}$$

Non-homogenous eqn

$\rightarrow$ constant

$$\therefore \boxed{v_p(x) = A = \delta}$$

$$\therefore \boxed{v(x) = C_1 \cos kx + C_2 \sin kx + \delta} \rightarrow (4) \quad (6)$$

Full solution

Boundary conditions,

$$(i) \quad \boxed{v(x=0) = 0} \quad \text{From (4)}$$

$$\Rightarrow v(0) = C_1 + C_2(0) + \delta = 0$$

$$\Rightarrow \boxed{C_1 = -\delta} \rightarrow (5)$$

$$(ii) \quad \frac{dv(x=0)}{dx} = 0 \Rightarrow -kC_1 \sin(0) + kC_2 \cos(0) + \frac{d\delta}{dx} = 0$$

$$\Rightarrow kC_2 = 0 \Rightarrow \boxed{C_2 = 0} \rightarrow (6)$$

From (4), (5) & (6)

$$v(x) = -\delta \cos kx + \delta$$

$$\Rightarrow \boxed{v(x) = \delta(1 - \cos kx)} \rightarrow (7)$$

Recall: $\delta \equiv v(x=L)$

$\therefore$ Eqn (7) becomes

$$\boxed{\delta = \delta(1 - \cos kL)} \rightarrow (8)$$

Eqn (8) is true when $\cos kL = 0 \quad \forall k$

$$\therefore \boxed{kL = n \frac{\pi}{2}} \quad \text{where: } n = 1, 3, 5, \dots$$

$$\Rightarrow \boxed{kL = \sqrt{\frac{P}{EI_{zz}}} \quad L = \frac{n\pi}{2} \rightarrow (9)$$

$$\left\{ \therefore k^2 = \frac{P}{EI_{zz}} \right\}$$

(7)

$$\Rightarrow \sqrt{\frac{P}{EI_{22}}} L = \frac{n\pi}{2}$$

$$\Rightarrow \frac{P}{EI_{22}} L^2 = \frac{n^2 \pi^2}{4}$$

$$\Rightarrow \boxed{P = \frac{n^2 \pi^2}{4L^2} EI_{22}} \quad \text{--- (10)}$$

Material stiffness $\rightarrow$

Buckling load or (Axial compressive load where we have buckling) $\rightarrow$

(Length of bone) $\rightarrow$

(Second moment of area) $\rightarrow$

Smallest Buckling load or critical Buckling Load:

$$\boxed{n=1 \text{ in Eqn (10)}}$$

$$\Rightarrow \boxed{P_{cr} = \frac{\pi^2}{4L^2} EI_{22}} \quad \text{--- (11)}$$

Critical Buckling load
for a cantilevered column